National University of Computer and Emerging Sciences - FAST

CS303- Software Engineering

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SECTION A – Case Studies: Attempt all questions (40 Marks)

Case Study I (20 marks)

The Election Commission of Pakistan has decided to acquire a Turnkey Solution for Election Management System. The new computerized system will be based on a real time online Electronic Voting System that will replace the traditional and conventional system of vote casting through ballots.

The rules and regulations have to be modified as the new system will break the geographical and location barriers as vote casting can be accomplished by an individual voter anywhere in the country or worldwide. The voters will just have to cast the votes through Biometric Devices (Fingerprint Recognition System) at the permanent/temporarily set computerized Vote Casting Facilities or Mobile Units. The infrastructure of computerized vote casting system can be applied to worldwide Pakistani voters with online vote casting center setups. The Elections results are to be available on Election Commission Website and also an integral part of the project. The Election Commission has already a quality website online but without the Election Results Facility.

The **software solution project** has been outsourced to Extensible Technologies (PVT) Limited. The software company has the exposure of handling large projects at international level with similar nature of expertise. As a Software Project Manager at Extendible Technologies analyzes the given project scenario and answer the following:

Part a. (10 marks total)

1. Identify the stakeholders according to the specific types for the achievement of the project. Identify and establish objectives and sub-objectives for each type of stakeholders. For each objective identify measures of effectiveness. Identify the project scope
2. Prepare a questionnaire to evaluate the software requirements.
3. Prepare a list of requirements of both types.

Part b. Prepare a Scope Document. Write for the following contents. (10 marks)

1. **Business Requirements**

1.1. Background, Business Opportunity, and Customer Needs

1.2. Business Objectives and Success Criteria

1.3. Business Risks

2. Vision of the Solution

2.1. Vision Statement

2.2. Major Features

2.3. Assumptions and Dependencies

3. Business Context (Application)

Case Study II (10 Marks)

"An automated ticket issuing system sells rail tickets. User selects their destination and input a credit card and a personal identification number. The rail ticket is issues and their credit card account charged. When the user presses the start button, a menu display of potential destinations is activate, along with a message to the user to select a destination. Once a destination has been selected, users are requested to input their credit card. Its validity is checked and the use is than requested to input a personal identifier. When the credit transaction has been validated, the ticket is issued"

Use above case study for Question ~~1 to 4~~ following

Q15. List 3 core system requirements (2 marks)
Q16. List down all use cases and then draw all of them (5 Marks)
Q17. If Spiral is selected as a process model, how will you as a project manager are making project plan, resource requirements and project delivery plans? (3 Marks)

Case Study III (10 Marks)

IDENTIFY FUNCTIONAL AND NON FUNCTIONAL REQ's
ABC, Inc. has experienced phenomenal growth over the last three years due to mergers and acquisitions of companies that are similar or complementary in nature. To enhance ABC's competitive position, management desires an HR system to identify employees with needed skills and training across all company locations. ABC management has determined that developing a new comprehensive HR system across the company would be cost prohibitive. Instead, management wishes to leave the legacy systems of each acquired company in place and integrate the data contained in them into an IT framework. The framework will consist of an IT infrastructure at headquarters, the public telephone network, and communications equipment at each company location necessary to support the framework processes. The framework, code named SATURN (for Skills and Training Unified Referral Network), will not replace any of the legacy systems of the individual companies that ABC has acquired. The presentation of the new companywide employee data will be transparent to the local user. In other words, each local user will see the selected employee information for the whole company in the same format as it would be if stored locally on their own legacy system. In addition, SATURN will be able to determine whether all legacy systems are available at the time a query is made. Should a particular legacy system be unavailable for any reason, the user will be notified so that a query can be made at a later time. Each query should be complete and all available information returned to the user within one minute. The SATURN system should be able to support up to 20 concurrent users without any degradation of performance.

SECTION B – Short Q/A: Attempt all questions with appropriate examples (30 Marks-3 marks each)
Q1. You have been given the responsibility of eliciting requirements from four different customers, some of whom are too busy to meet you. What should you do?
Q2. Are there ever circumstances where you can actually skip requirements gathering?
Q3. How is Functional Independence related to Cohesion and Coupling?
Q4. What is System Testing and each of the 4 tests included in System Testing.
Q5. Why does an iterative process make it easier to manage?
Q6. Is every agile process necessarily iterative? Justify with examples.
Q7. Is it possible for a process to complete a project in just one iteration and still be agile? Explain your answers
Q8. What are decision criteria to adopt a design pattern?
Q9. List down the design pattern catalogues. (Hint: Design pattern catalogue is defined by)
Q10. Categorize from below which are for functional and which are for non-functional:

- Time/ speed
- Platform independency
- Space/ size
- Muti-User Access
- Portable
- Form submission
- Ease of use/ usability
- Data Security

- Reliability
- Access Authorization
- Portability

SECTION C – Descriptive: Attempt all questions (30 Marks - 6 marks each)

Q1. Describe in your own words the difference between Verification and Validation. Why are these tests necessary?

Q2. What does "win-win" mean in the context of negotiation during requirement engineering?

Q3. Should the Interface Analysis be completed prior to Interface Designing? What are the Interface design Steps?

Q4. Can Abstraction and Refinement complement each other? Explain

Q5. What is implied by 'Prescriptive Process Models' Name a few Prescriptive Process Models

**** GOOD LUCK ****